

Bar Charts

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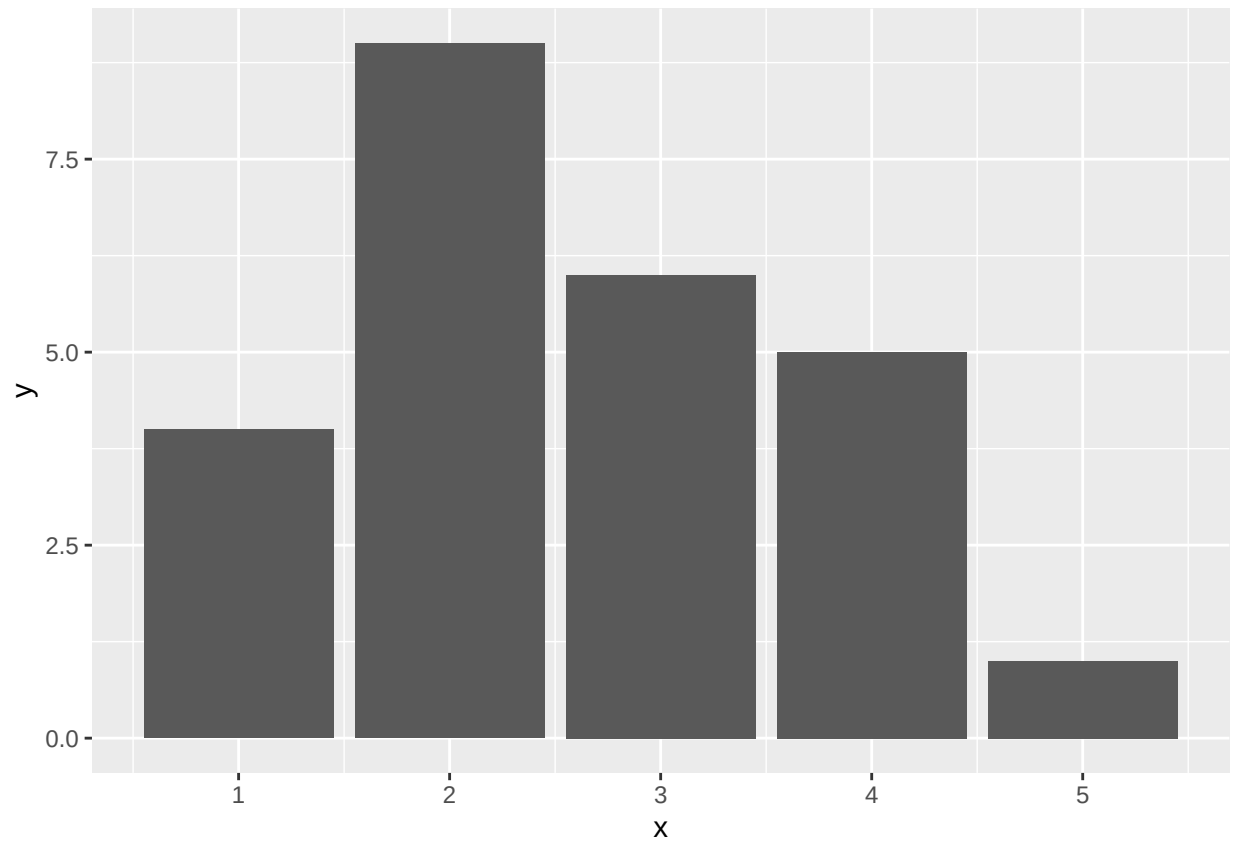
Bar Chart with Table

Discrete Variable

```
x <- 1:5  
y <- c(4, 9, 6, 5, 1)  
z <- data.frame(x, y)  
z
```

```
##   x y  
## 1 1 4  
## 2 2 9  
## 3 3 6  
## 4 4 5  
## 5 5 1
```

```
ggplot(data = z, aes(x = x, y = y)) + geom_bar(stat = "identity")
```

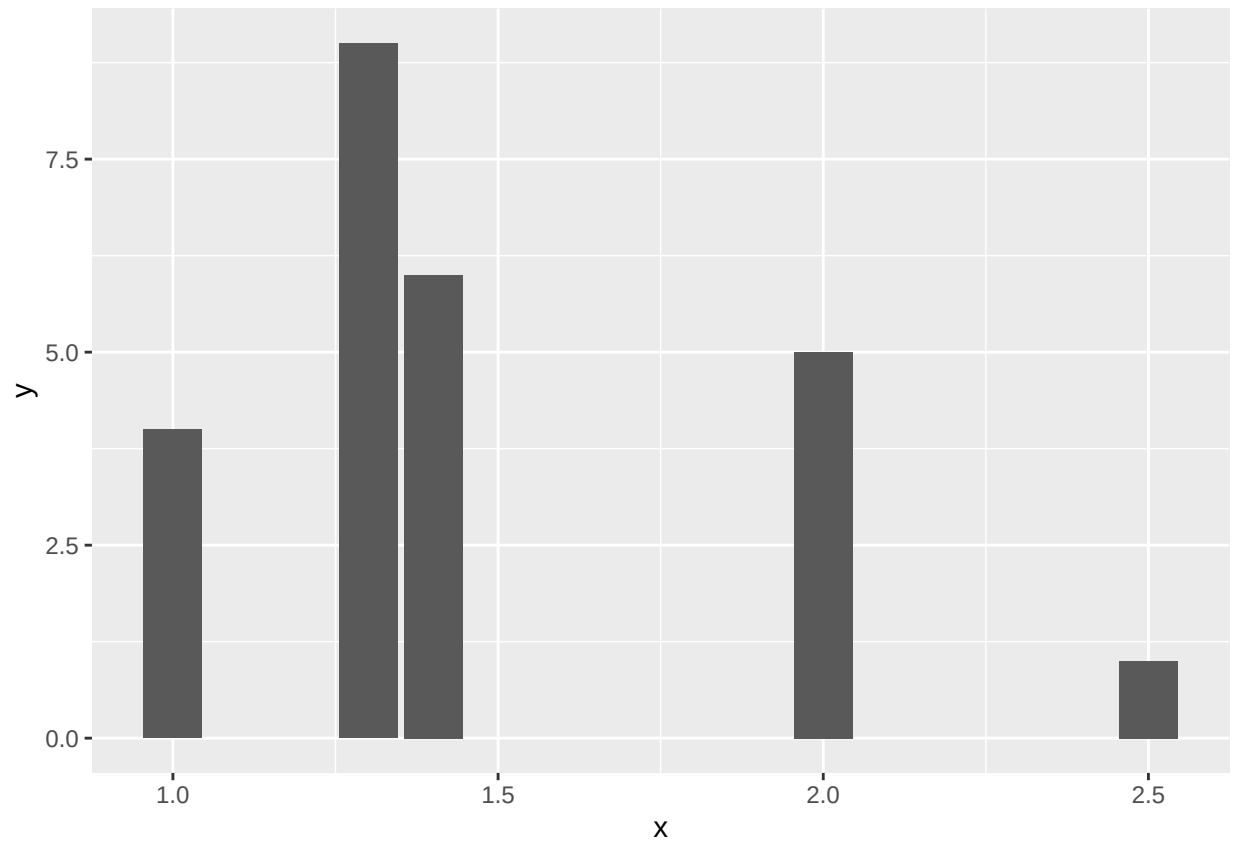


Continuous Variable

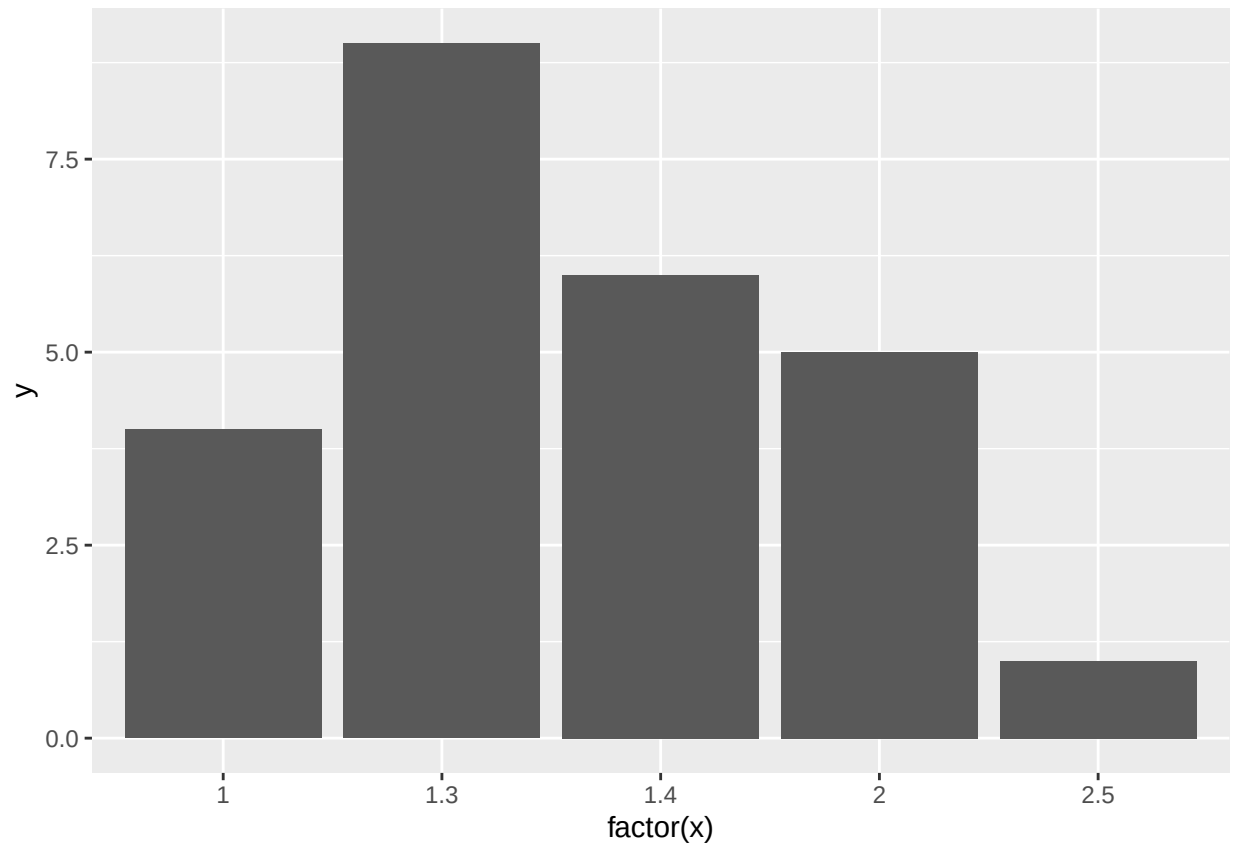
```
x <- c(1, 1.3, 1.4, 2.0, 2.5)
y <- c(4, 9, 6, 5, 1)
z <- data.frame(x, y)
z
```

```
##      x y
## 1 1.0 4
## 2 1.3 9
## 3 1.4 6
## 4 2.0 5
## 5 2.5 1
```

```
ggplot(data = z, aes(x = x, y = y)) + geom_bar(stat = "identity")
```

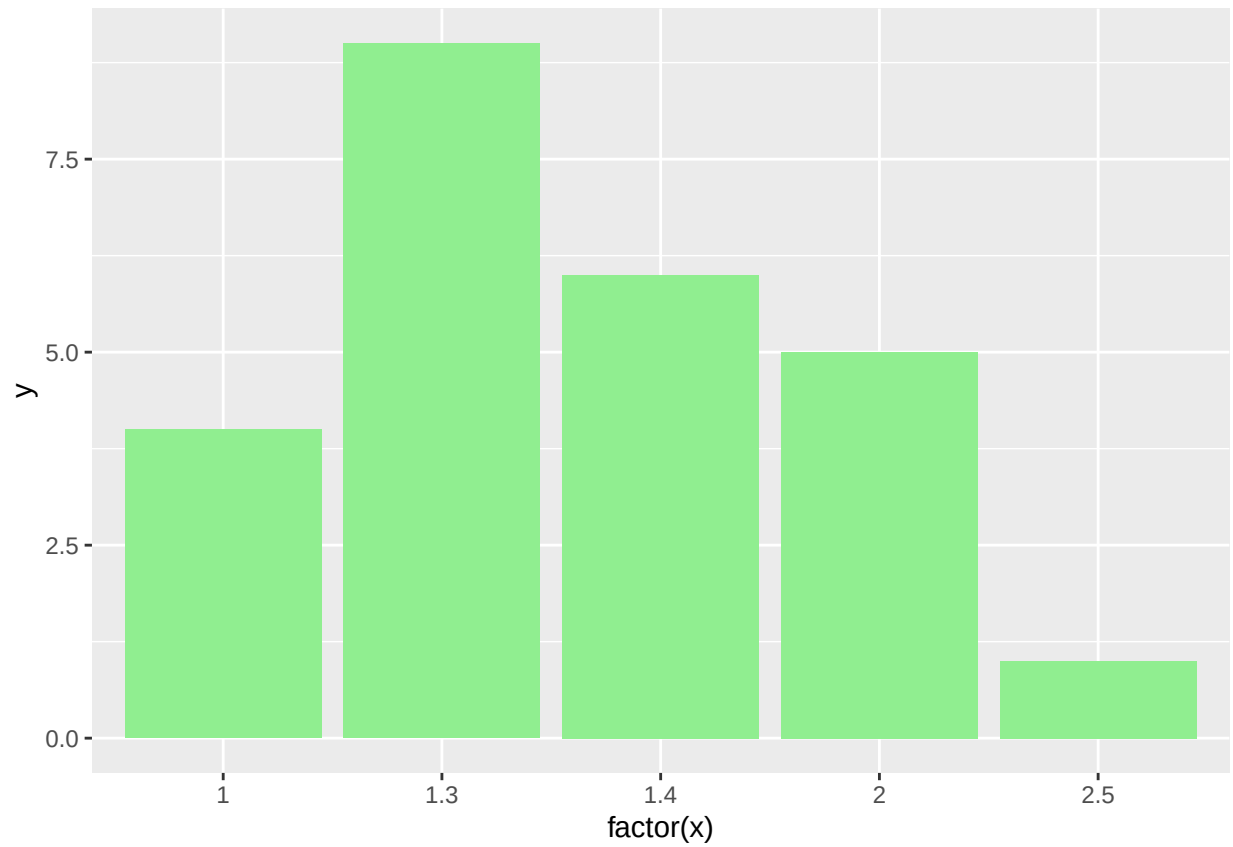


```
ggplot(data = z, aes(x = factor(x), y = y)) + geom_bar(stat = "identity")
```

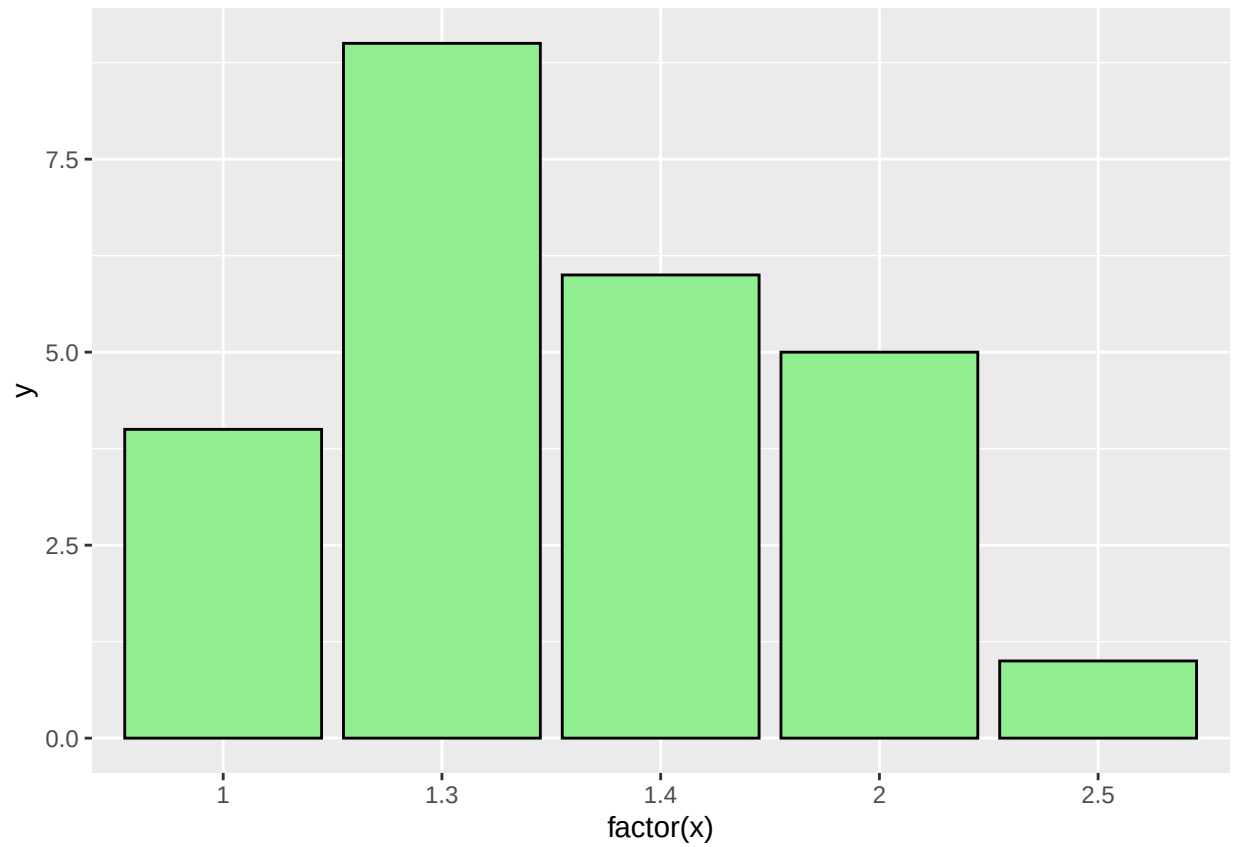


Change Colours and Add Border to Bars

```
ggplot(data = z, aes(x = factor(x), y = y)) + geom_bar(stat = "identity", fill = "lightgreen")
```



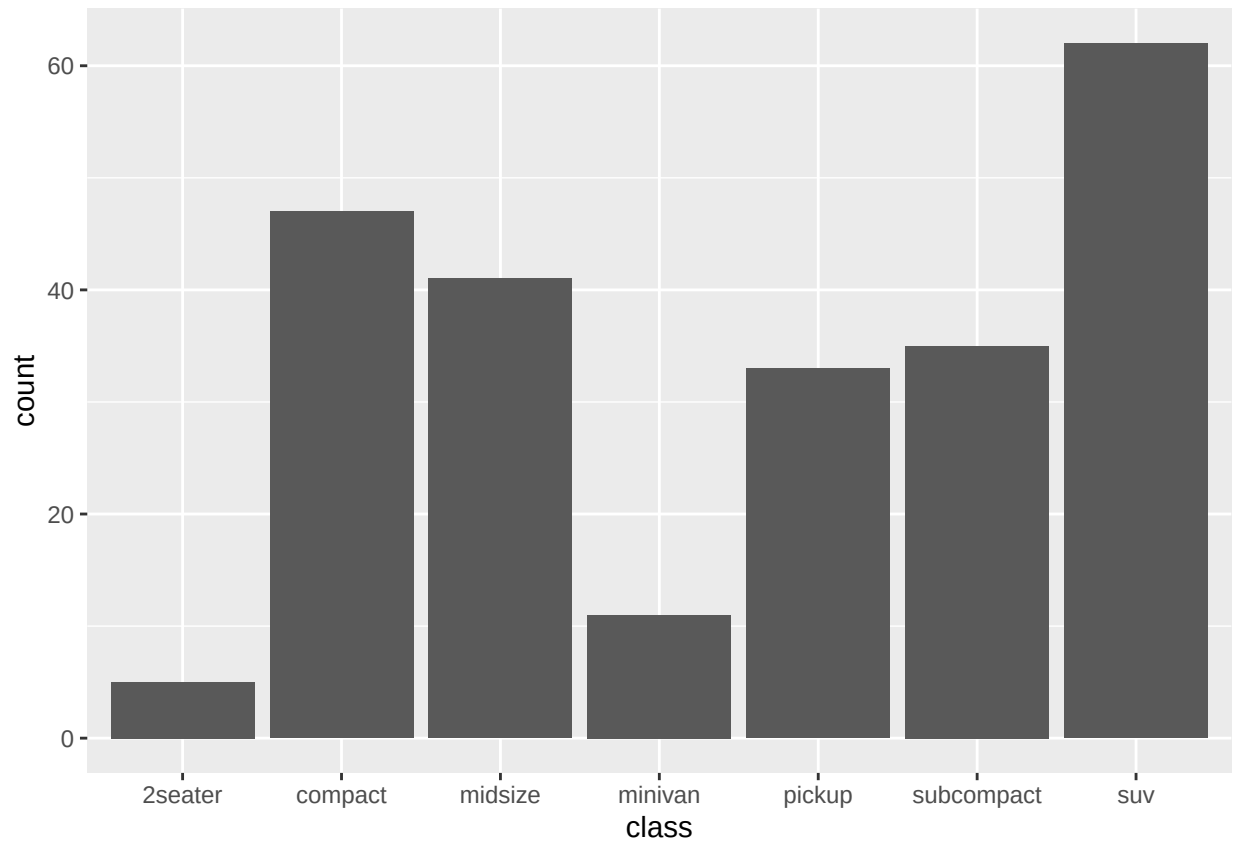
```
ggplot(data = z, aes(x = factor(x), y = y)) +  
  geom_bar(stat = "identity", fill = "lightgreen", colour = "black")
```



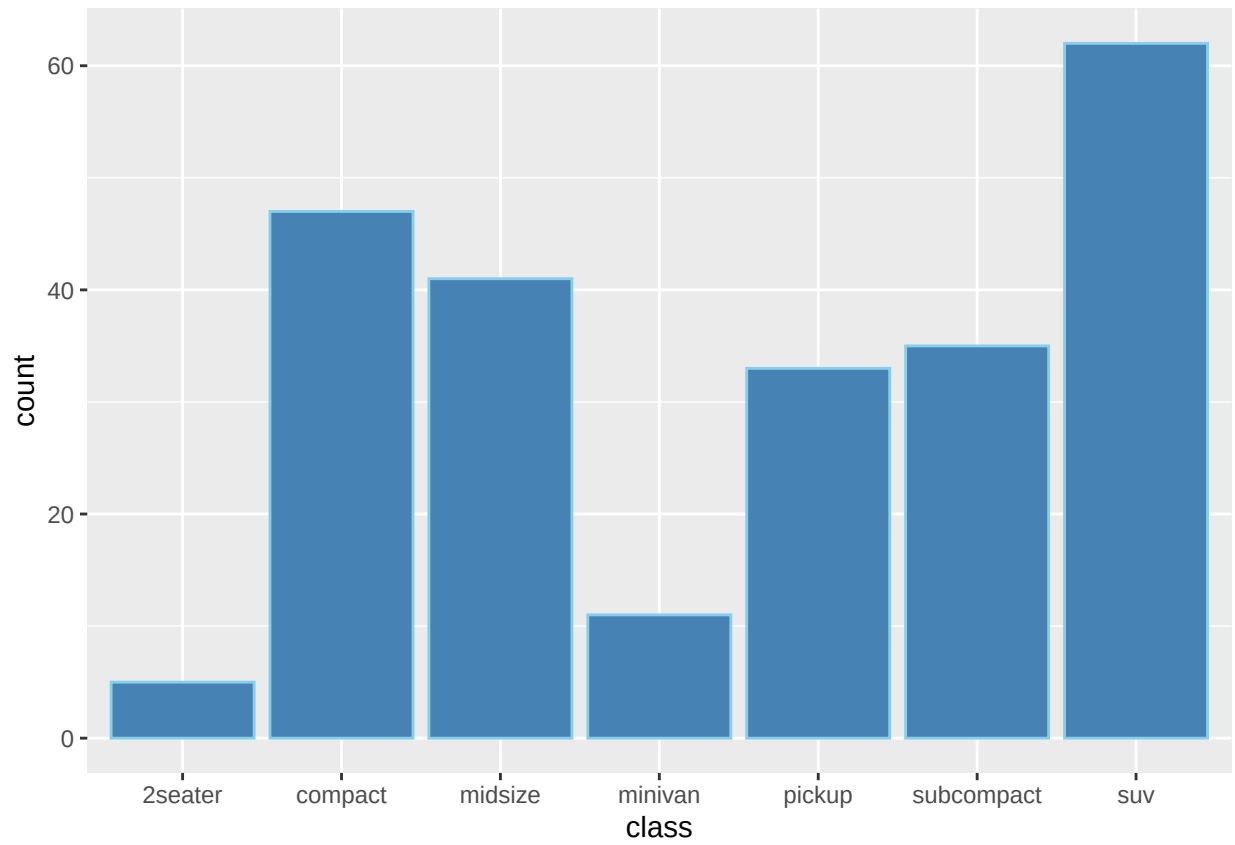
Bar Chart with Count Data

Discrete Variable

```
b <- ggplot(data = mpg)
b + geom_bar(aes(x = class))
```

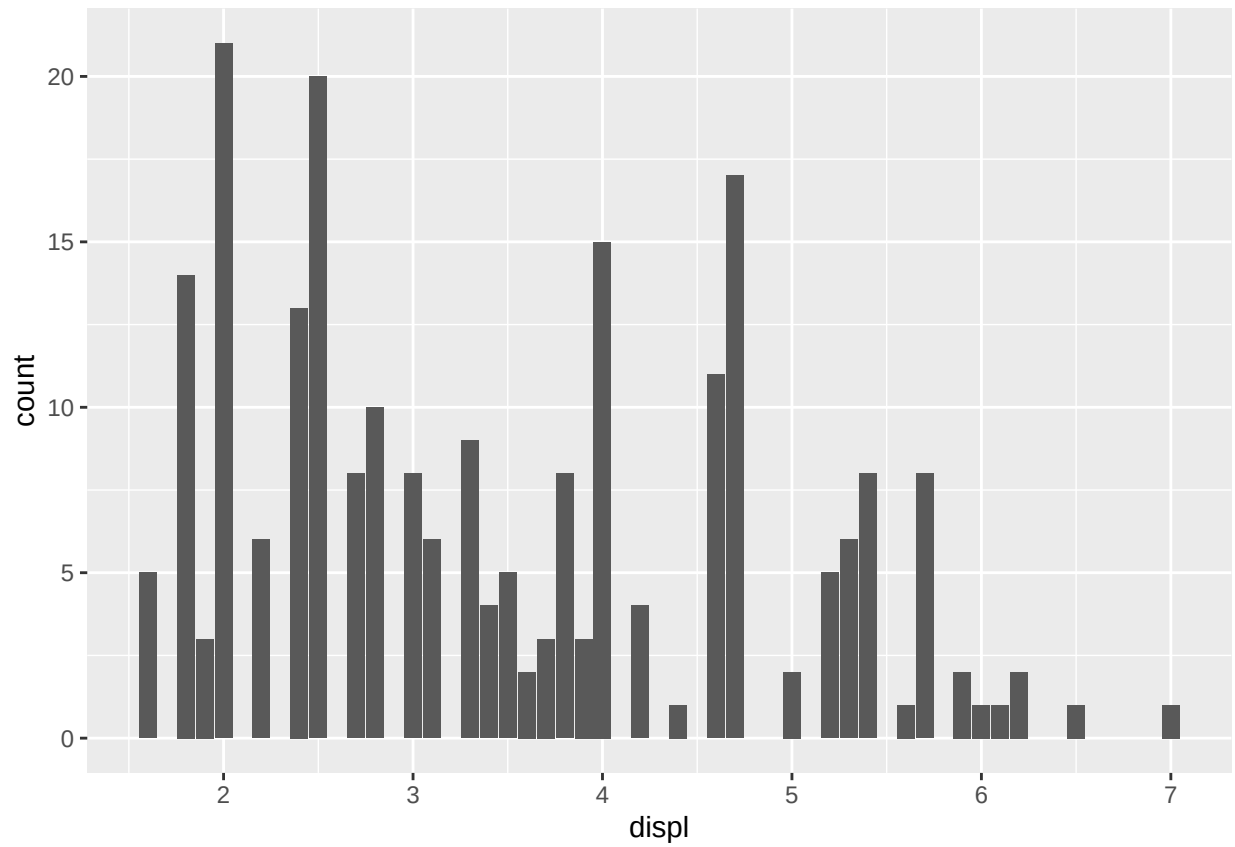


```
b + geom_bar(aes(x = class), colour = "skyblue", fill = "steelblue")
```

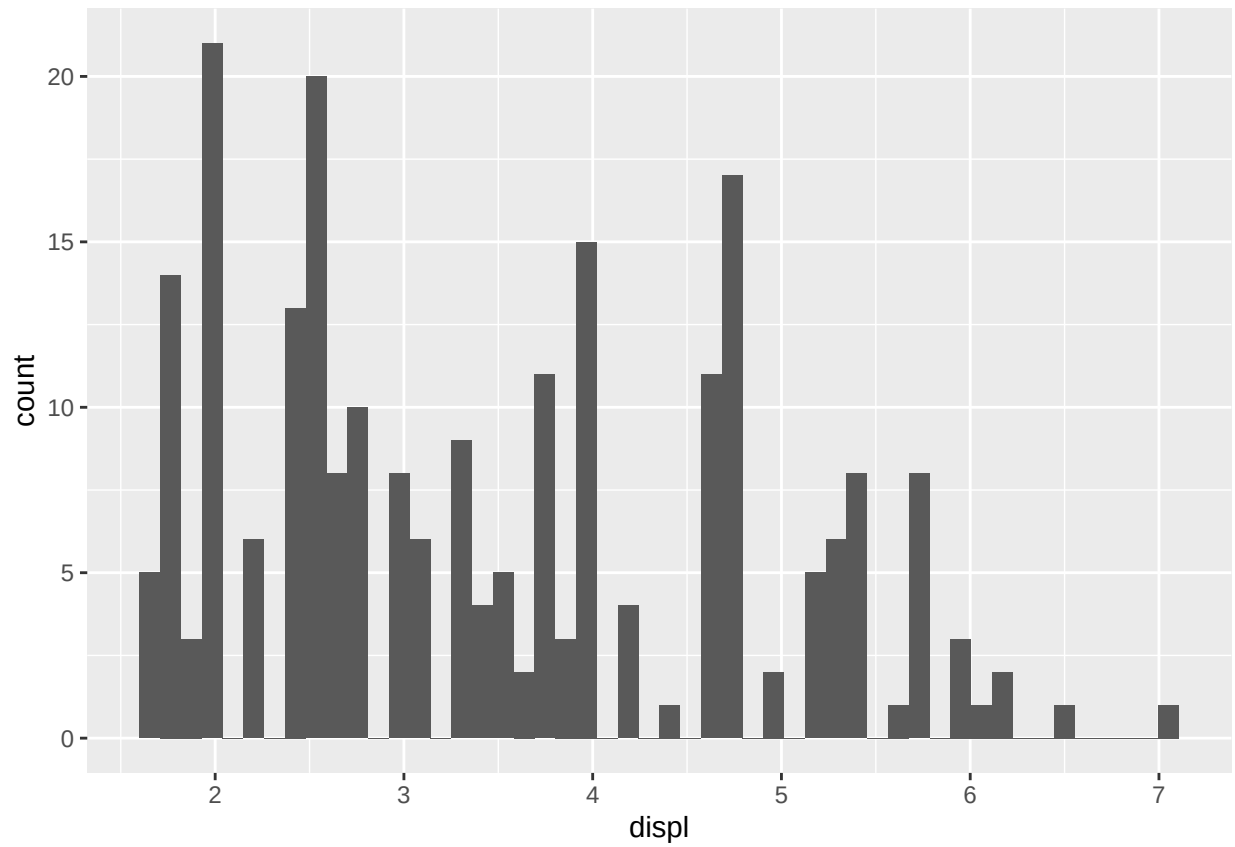


Continuous Variable

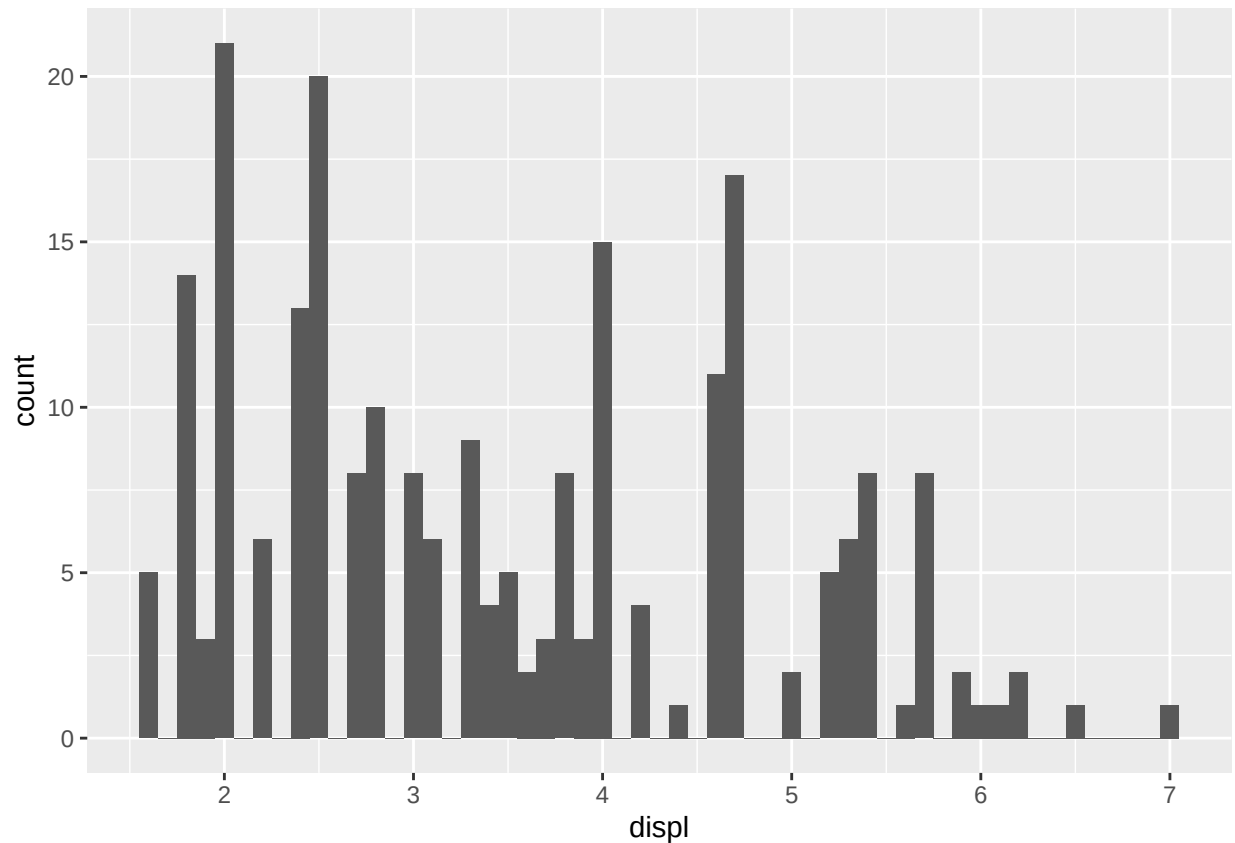
```
b <- ggplot(data = mpg)
b + geom_bar(aes(x = displ))
```

```
b + geom_histogram(aes(x = displ), bins = 50)
```



```
b + geom_histogram(aes(x = displ), binwidth = 0.1)
```

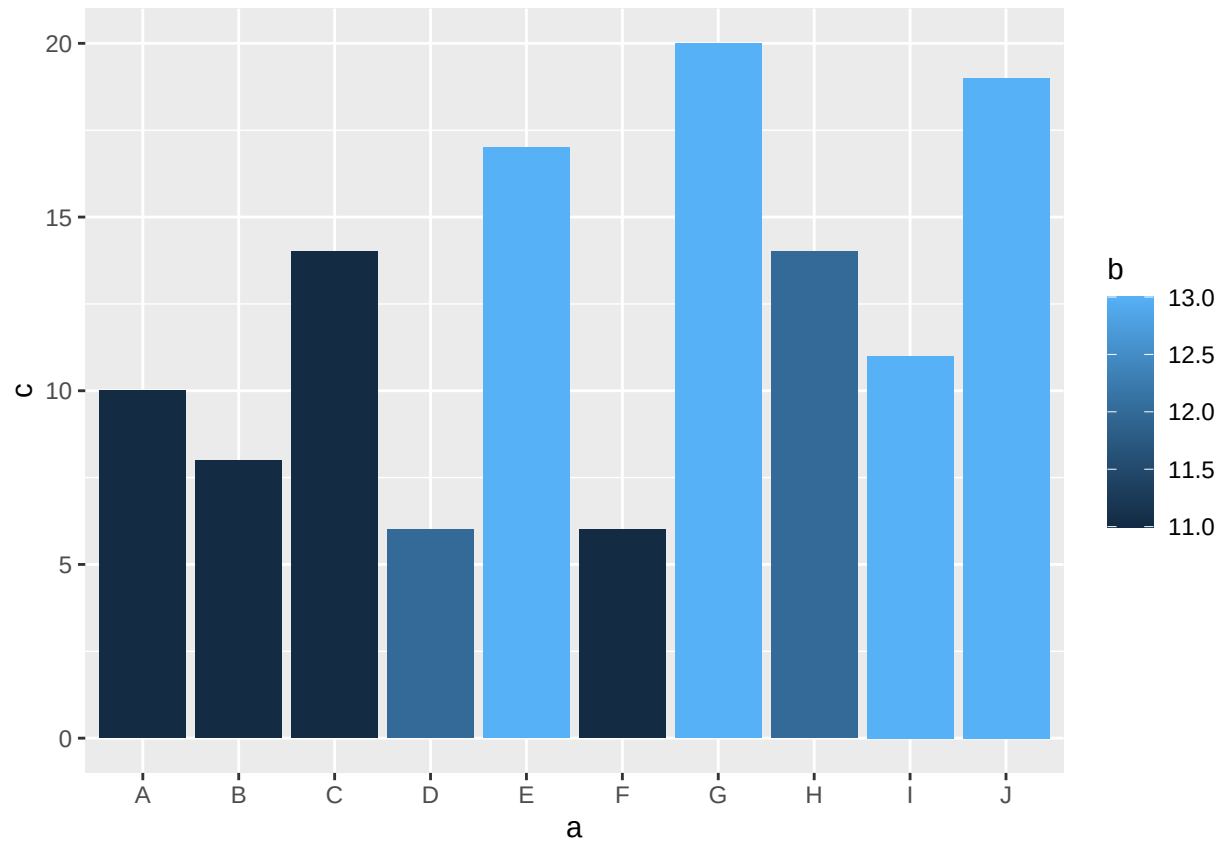


Change Colours with Category

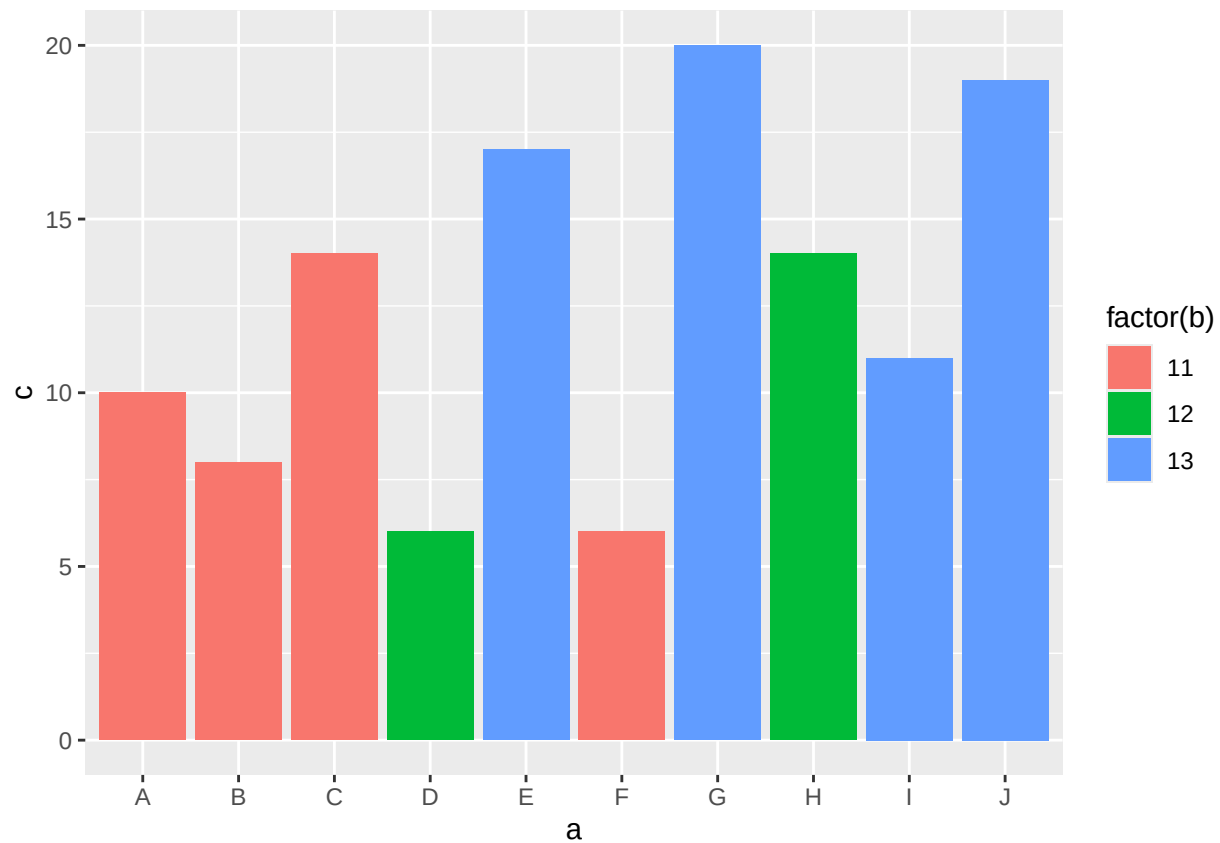
```
a <- c("A", "B", "C", "D", "E", "F", "G", "H", "I", "J")
b <- c(11, 11, 11, 12, 13, 11, 13, 12, 13, 13)
c <- c(10, 8, 14, 6, 17, 6, 20, 14, 11, 19)
d <- data.frame(a, b, c)
d
```

```
##   a  b  c
## 1  A 11 10
## 2  B 11  8
## 3  C 11 14
## 4  D 12  6
## 5  E 13 17
## 6  F 11  6
## 7  G 13 20
## 8  H 12 14
## 9  I 13 11
##10  J 13 19
```

```
ggplot(data = d, aes(x = a, y = c, fill = b)) + geom_bar(stat = "identity")
```



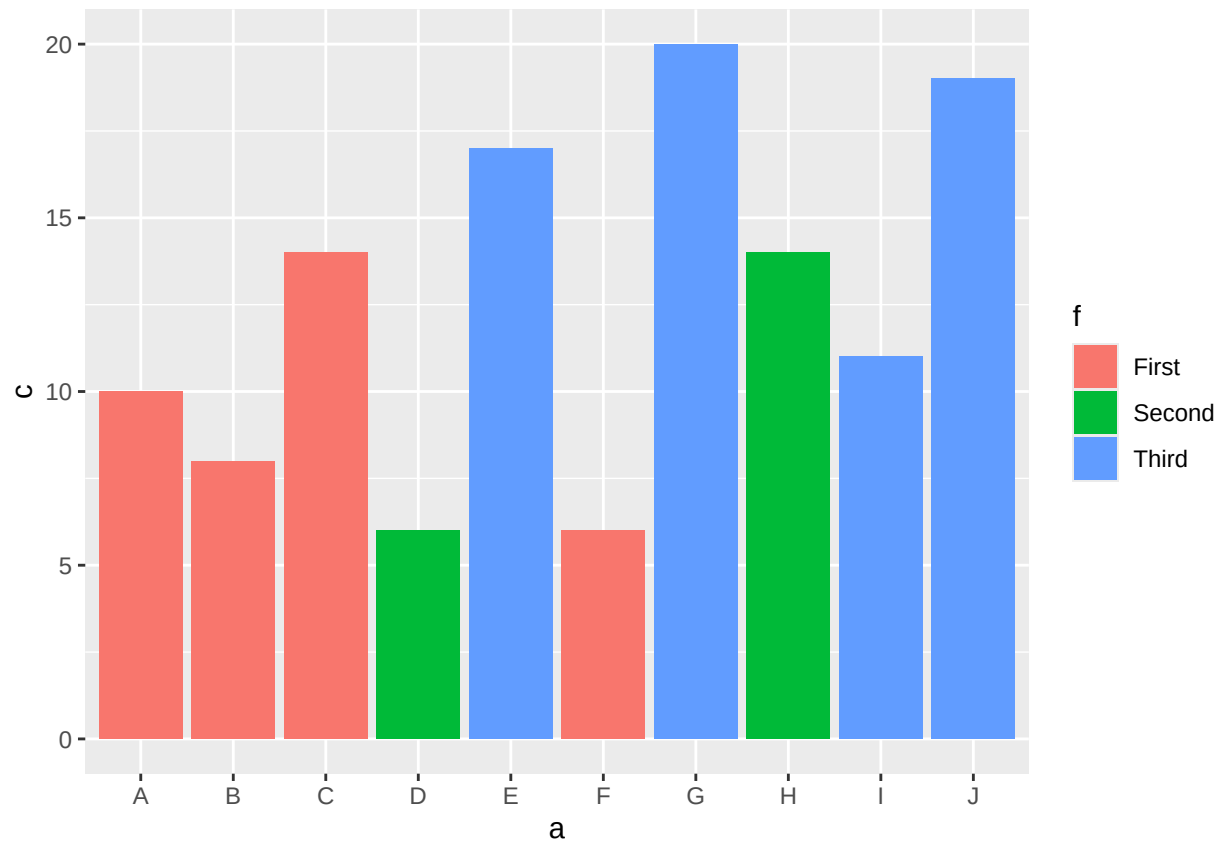
```
ggplot(data = d, aes(x = a, y = c, fill = factor(b))) + geom_bar(stat = "identity")
```



```
a <- c("A", "B", "C", "D", "E", "F", "G", "H", "I", "J")
b <- c(11, 11, 11, 12, 13, 11, 13, 12, 13, 13)
c <- c(10, 8, 14, 6, 17, 6, 20, 14, 11, 19)
d <- data.frame(a, b, c)
f <- factor(b, labels = c("First", "Second", "Third"))
e <- data.frame(a, f, c)
e
```

```
##   a      f  c
## 1 A  First 10
## 2 B  First  8
## 3 C  First 14
## 4 D Second  6
## 5 E  Third 17
## 6 F  First  6
## 7 G  Third 20
## 8 H Second 14
## 9 I  Third 11
##10 J  Third 19
```

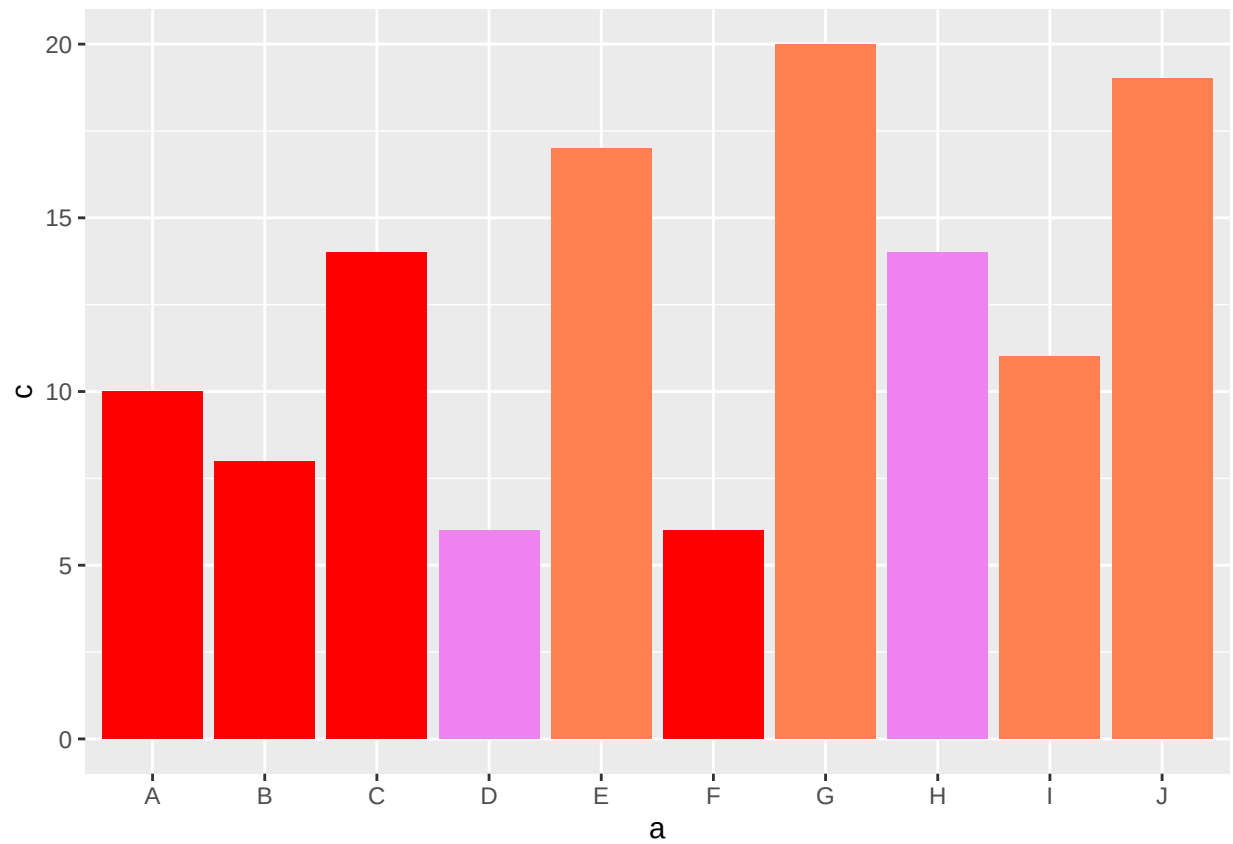
```
ggplot(data = e, aes(x = a, y = c, fill = f)) + geom_bar(stat = "identity")
```



```
g <- factor(b, labels = c("red", "violet", "coral"))
e <- data.frame(a, f, c, g)
e
```

```
##   a      f  c    g
## 1 A  First 10   red
## 2 B  First  8   red
## 3 C  First 14   red
## 4 D Second  6 violet
## 5 E  Third 17 coral
## 6 F  First  6   red
## 7 G  Third 20 coral
## 8 H Second 14 violet
## 9 I  Third 11 coral
##10 J  Third 19 coral
```

```
ggplot(data = e, aes(x = a, y = c)) + geom_bar(fill = g, stat = "identity")
```

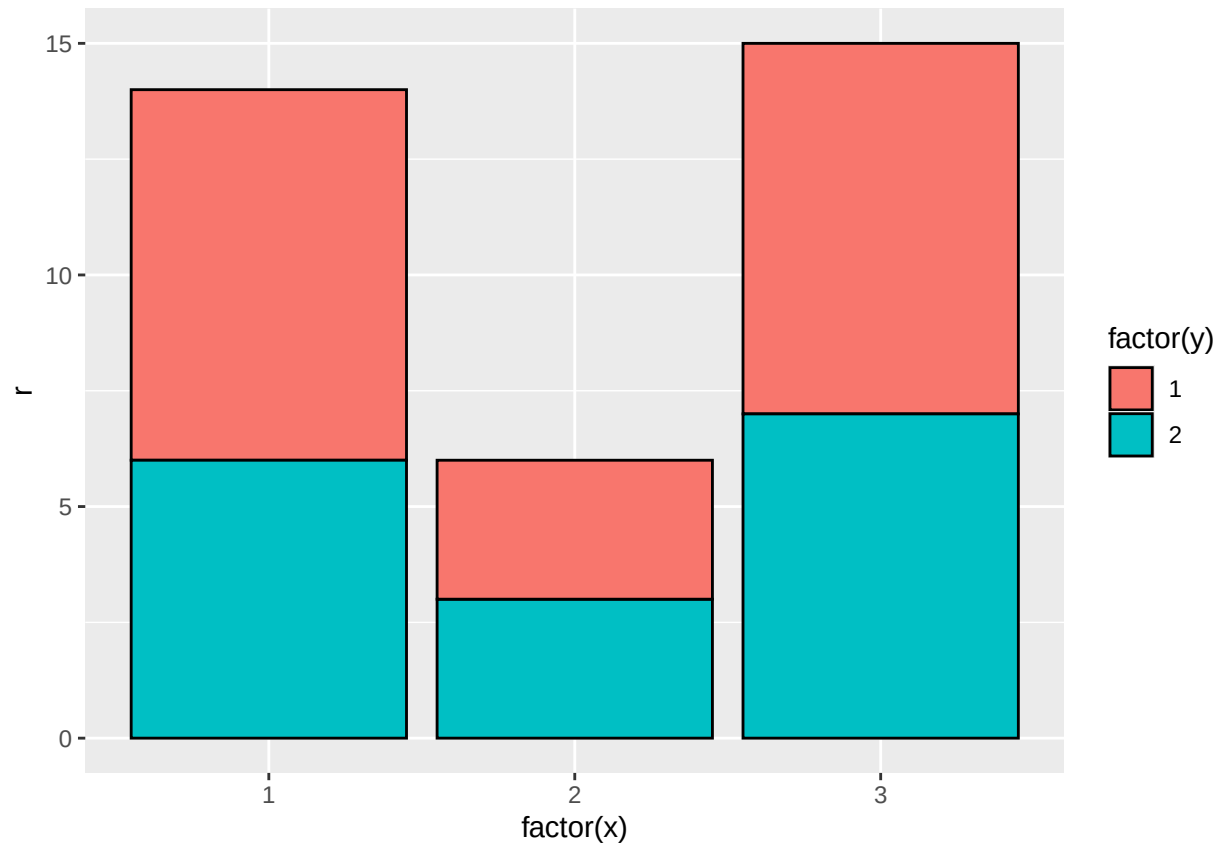


Group Data

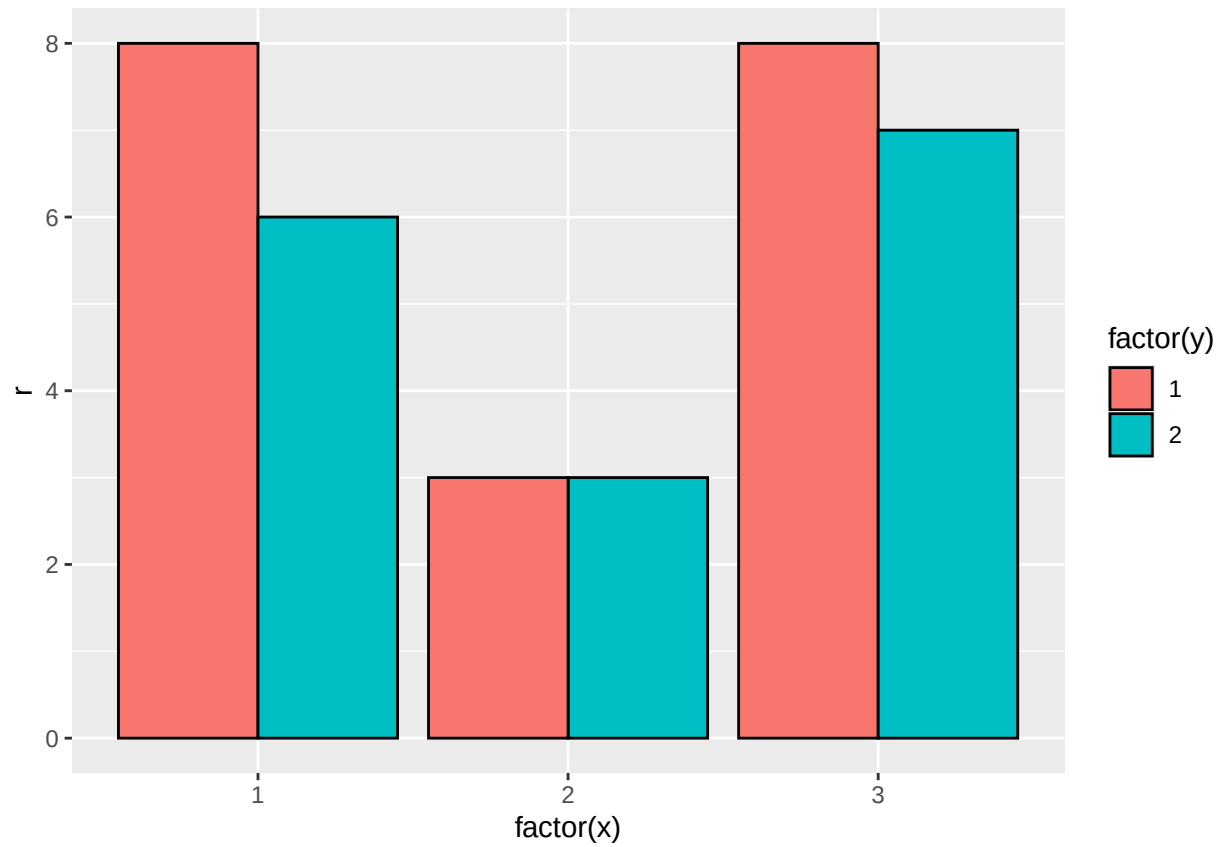
```
x <- rep(1:3, 2)
y <- rep(1:2, 3)
r <- c(8, 3, 8, 6, 3, 7)
z <- data.frame(x, y, r)
z
```

```
##   x y r
## 1 1 1 8
## 2 2 2 3
## 3 3 1 8
## 4 1 2 6
## 5 2 1 3
## 6 3 2 7
```

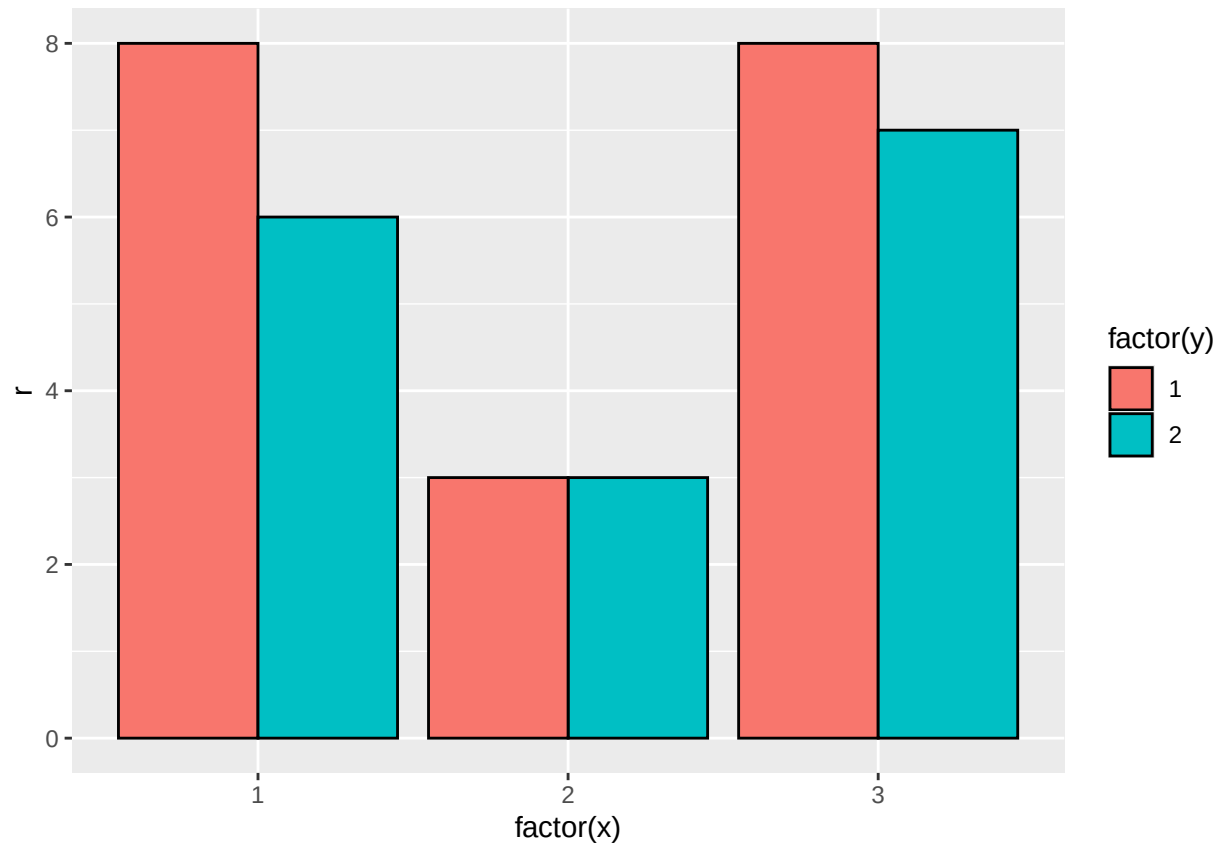
```
ggplot(data = z, aes(x = factor(x), y = r, fill = factor(y))) +
  geom_bar(stat = "identity", colour = "black")
```



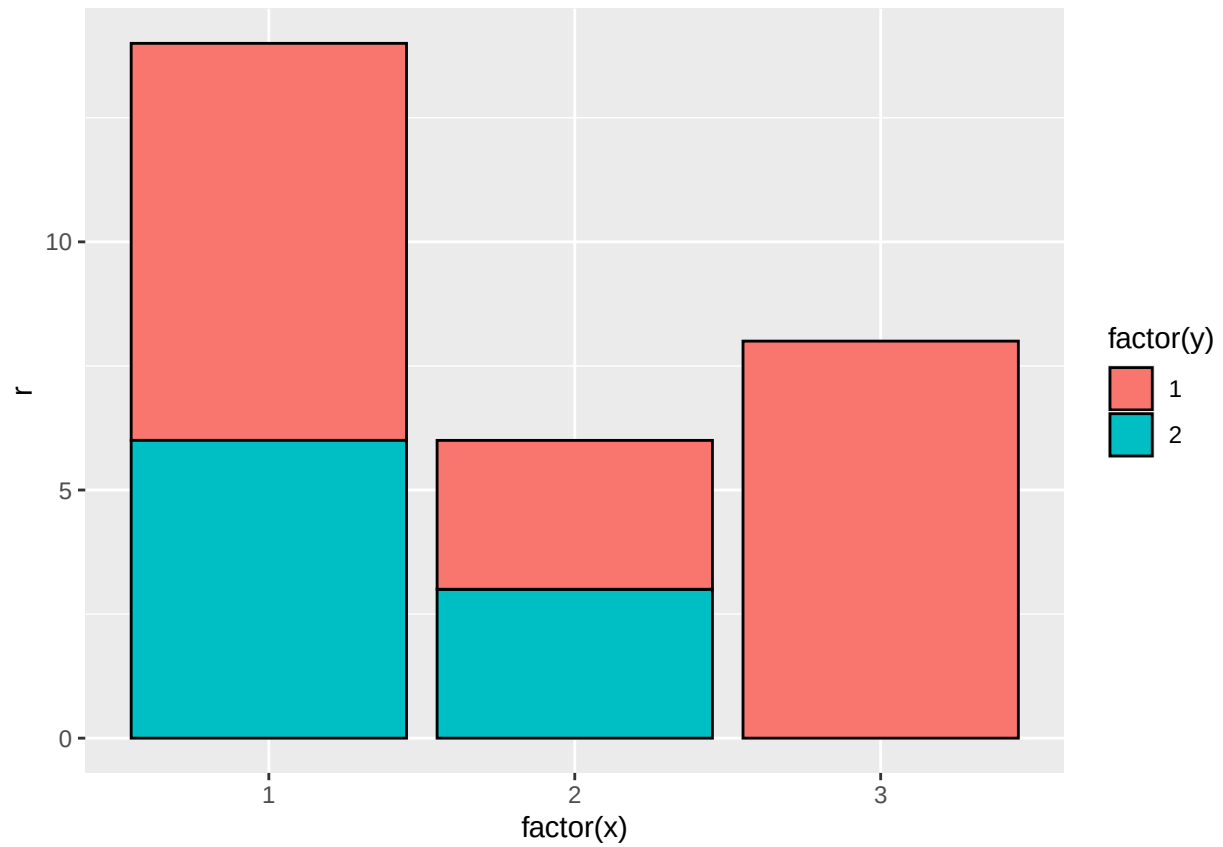
```
ggplot(data = z, aes(x = factor(x), y = r, fill = factor(y))) +  
  geom_bar(stat = "identity", colour = "black", position = "dodge")
```

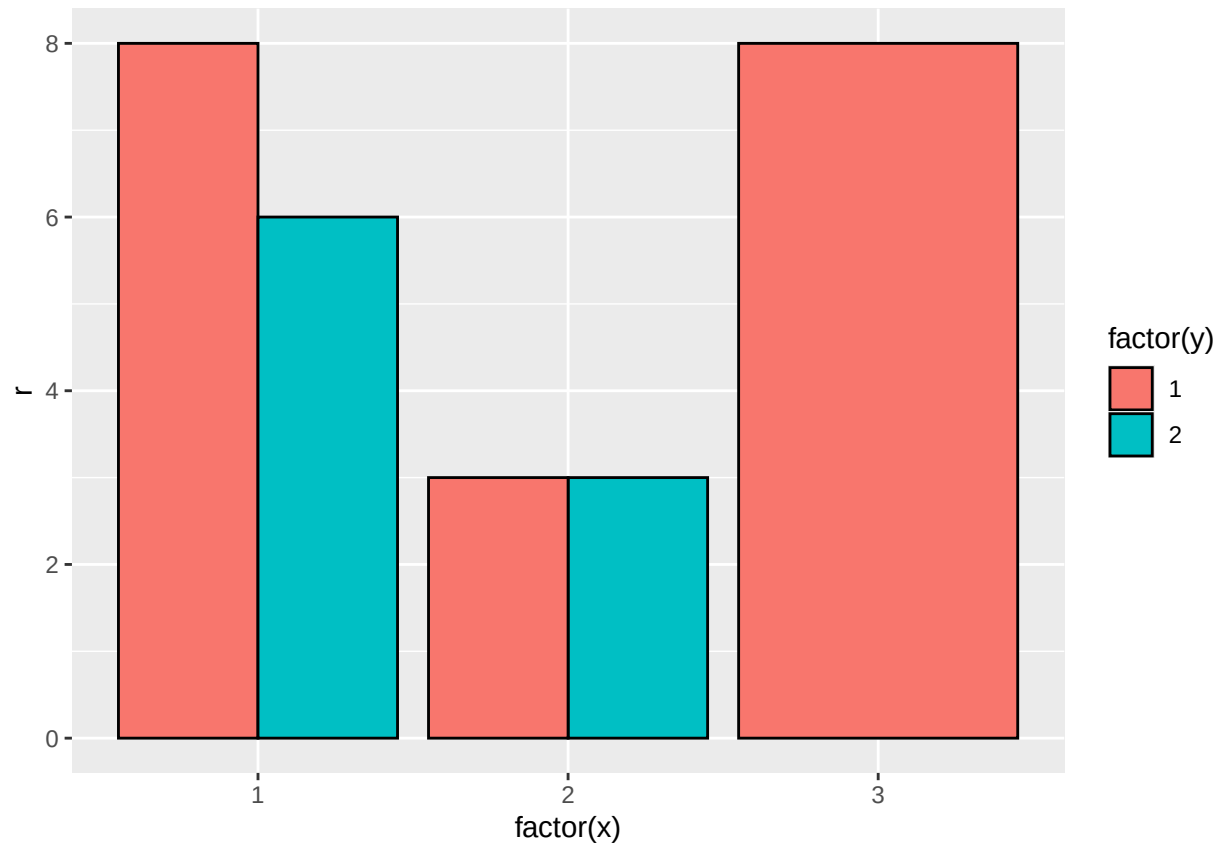
```
ggplot(data = z, aes(x = factor(x), y = r, fill = factor(y))) +  
  geom_bar(stat = "identity", colour = "black", position = position_dodge())
```



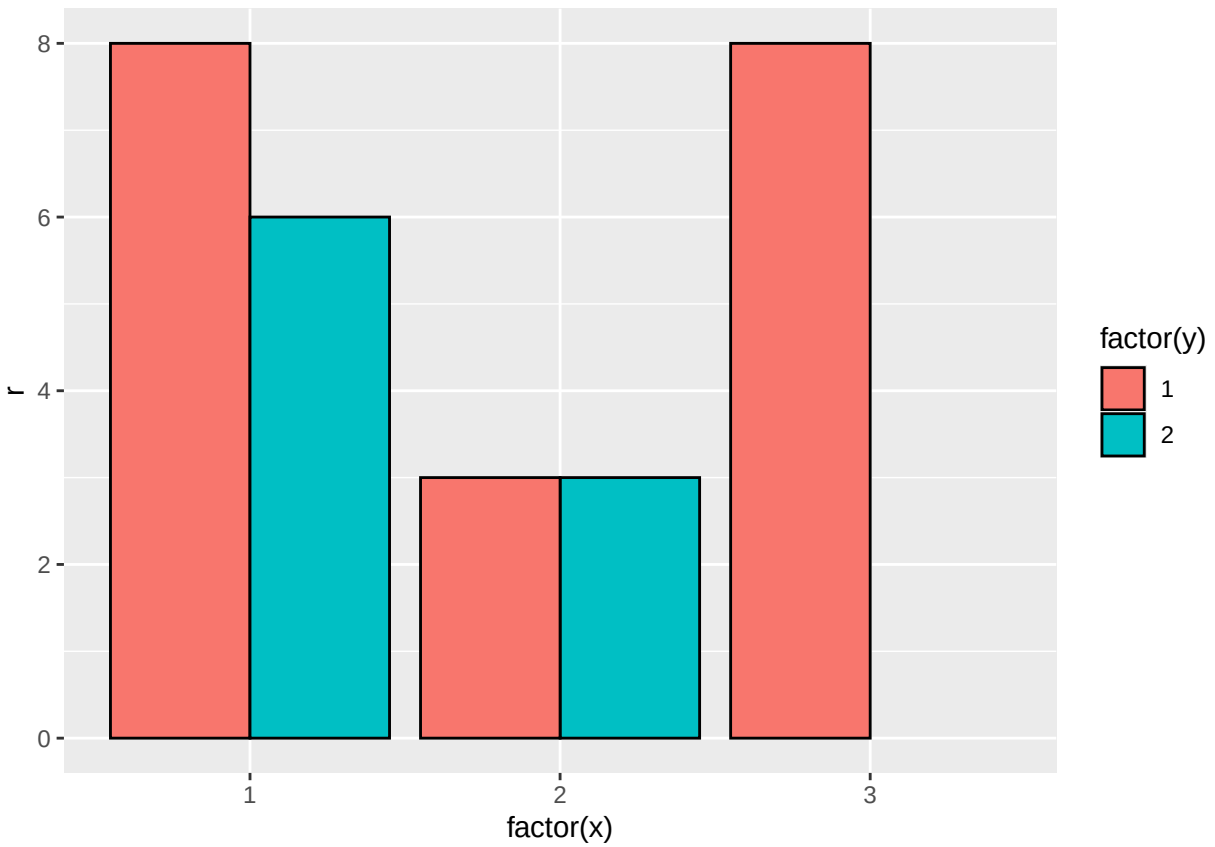
```
z1 <- z[-6, ]  
ggplot(data = z1, aes(x = factor(x), y = r, fill = factor(y))) +  
  geom_bar(stat = "identity", colour = "black")
```



```
ggplot(data = z1, aes(x = factor(x), y = r, fill = factor(y))) +  
  geom_bar(stat = "identity", colour = "black", position = "dodge")
```

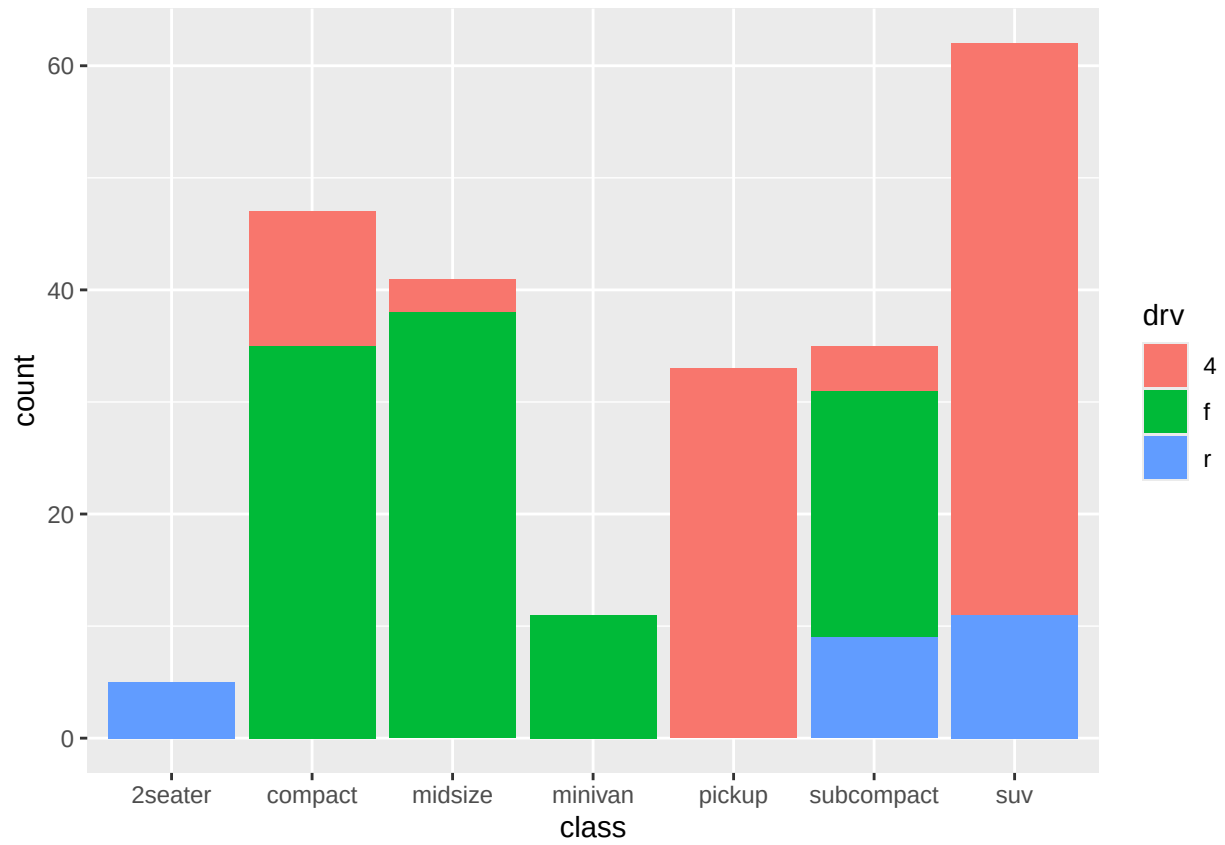


```
z2 <- rbind(z1, c(3, 2, NA))
ggplot(data = z2, aes(x = factor(x), y = r, fill = factor(y))) +
  geom_bar(stat = "identity", colour = "black", position = "dodge")
```

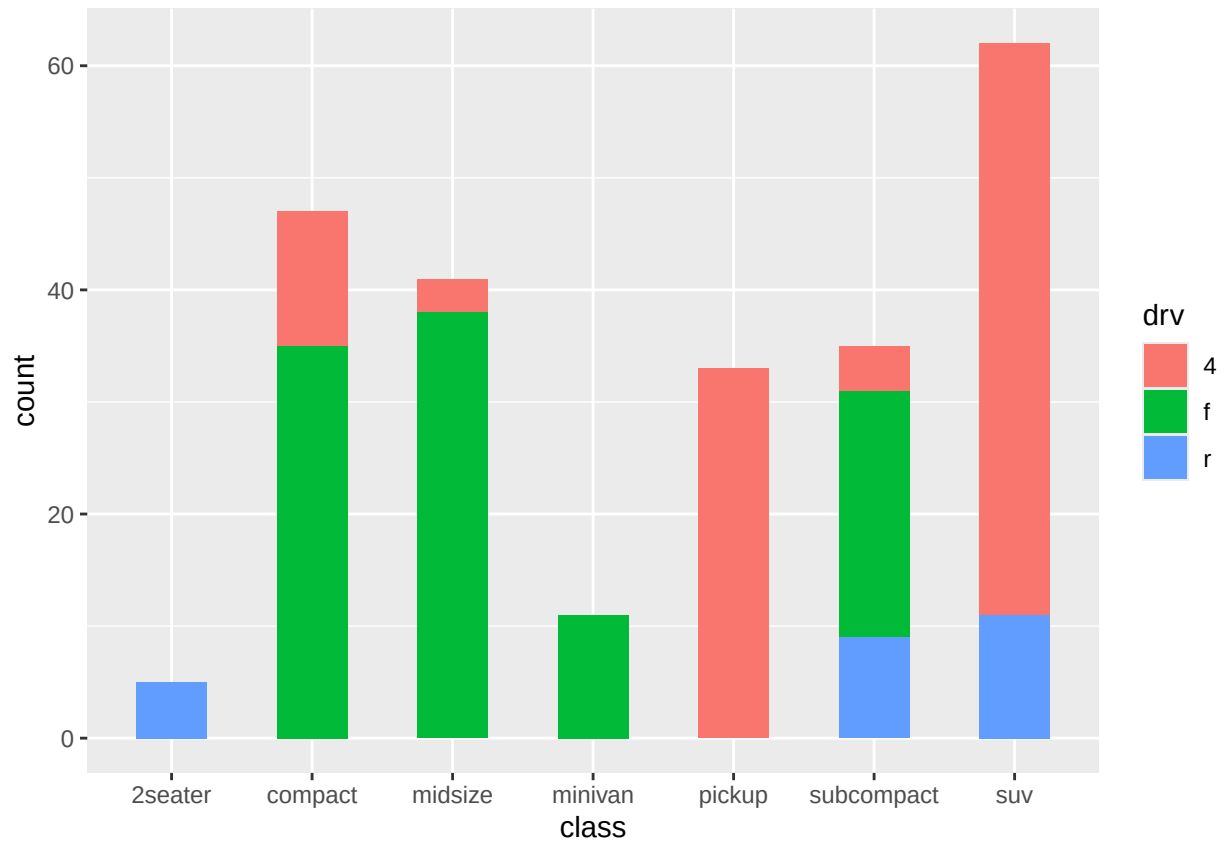


Set Bar Width and Spacing

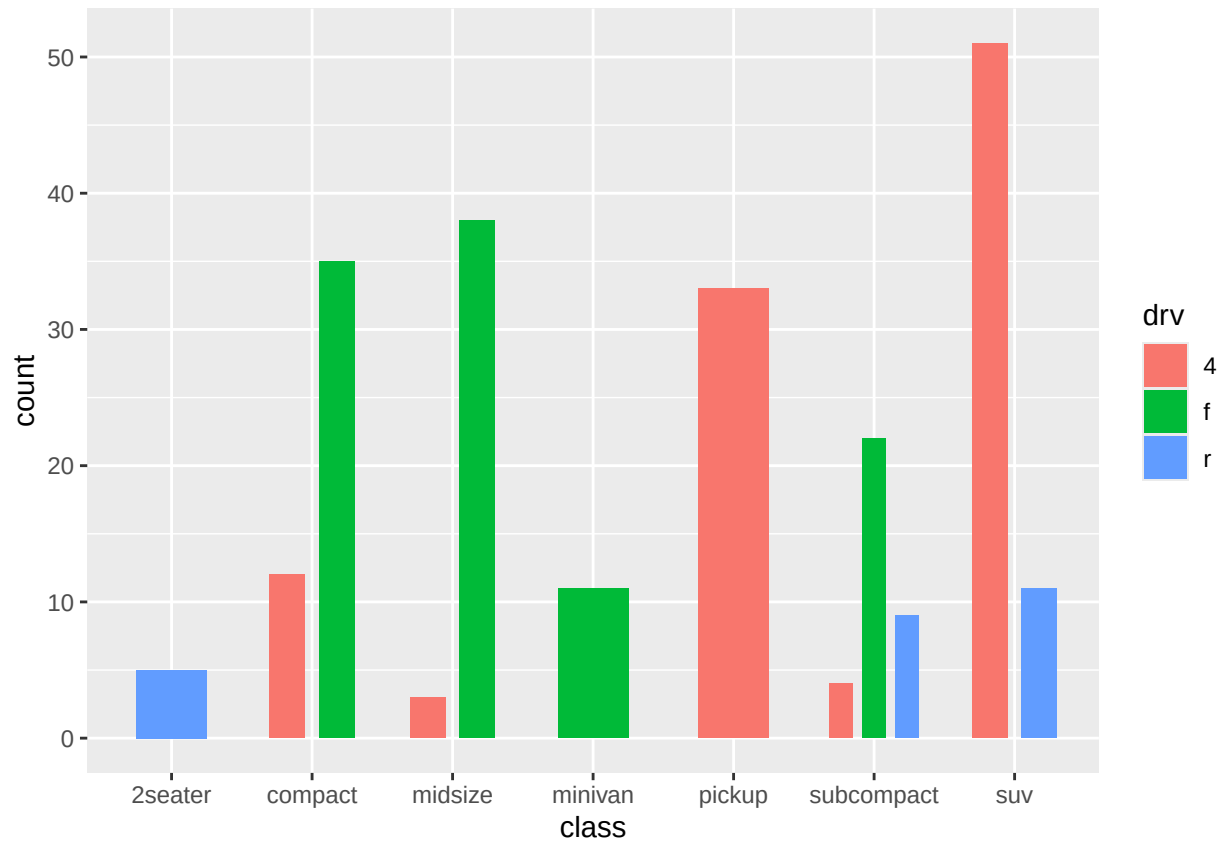
```
b <- ggplot(data = mpg, aes(x = class, fill = drv))  
b + geom_bar()
```



```
b + geom_bar(width = 0.5)
```



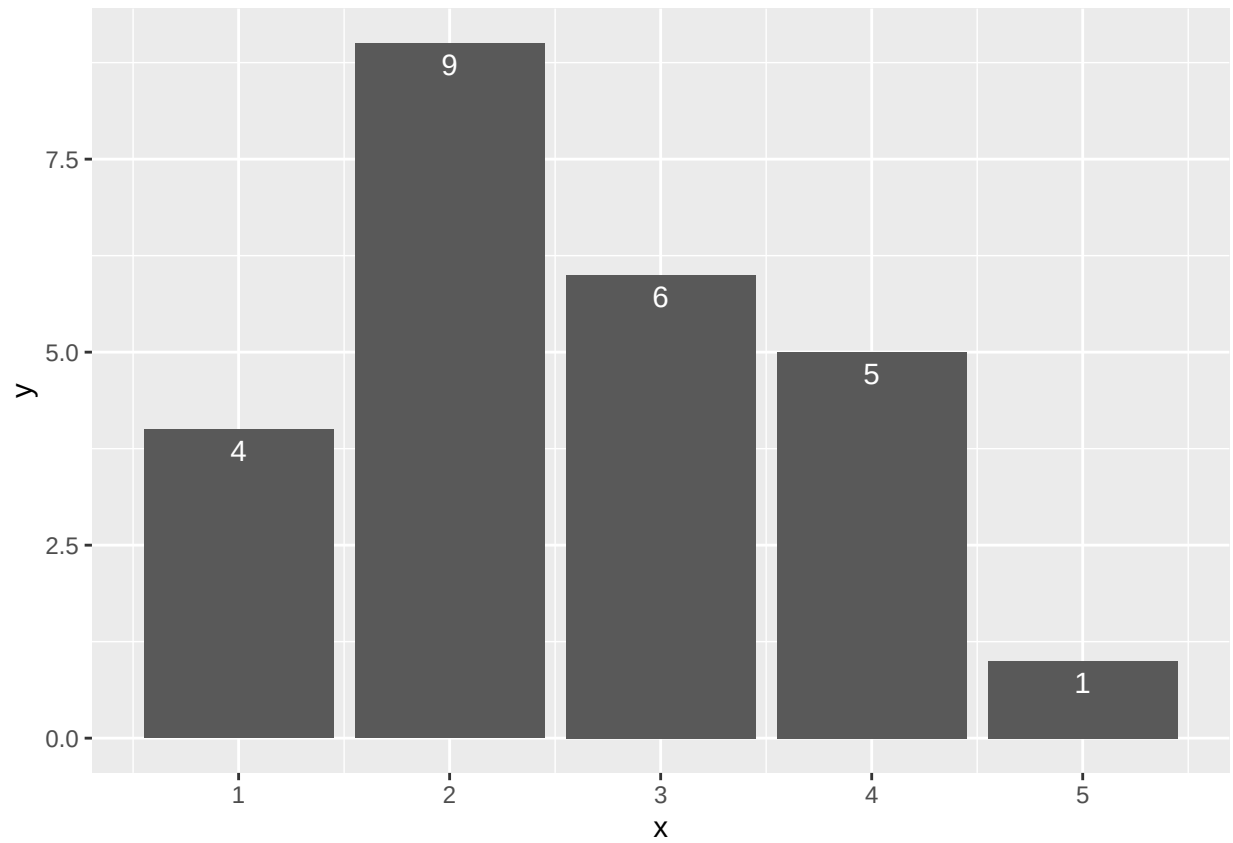
```
b + geom_bar(position = position_dodge(0.7), width = 0.5)
```



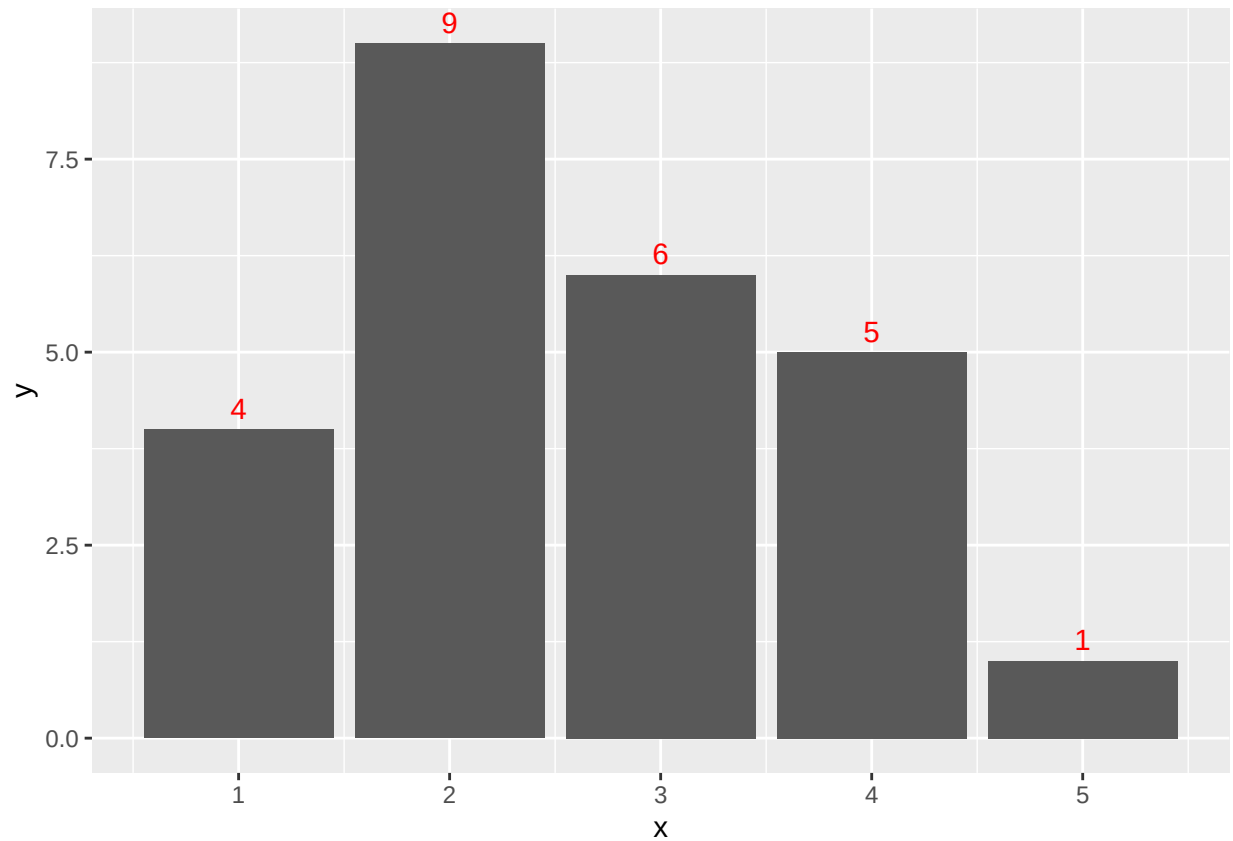
Add Labels

```
x <- 1:5
y <- c(4, 9, 6, 5, 1)
z <- data.frame(x, y)

ggplot(data = z, aes(x = x, y = y)) + geom_bar(stat = "identity") +
  geom_text(aes(label = y), vjust = 1.5, colour = "white")
```

```
ggplot(data = z, aes(x = x, y = y)) + geom_bar(stat = "identity") +  
  geom_text(aes(label = y), vjust = -0.5, colour = "red")
```

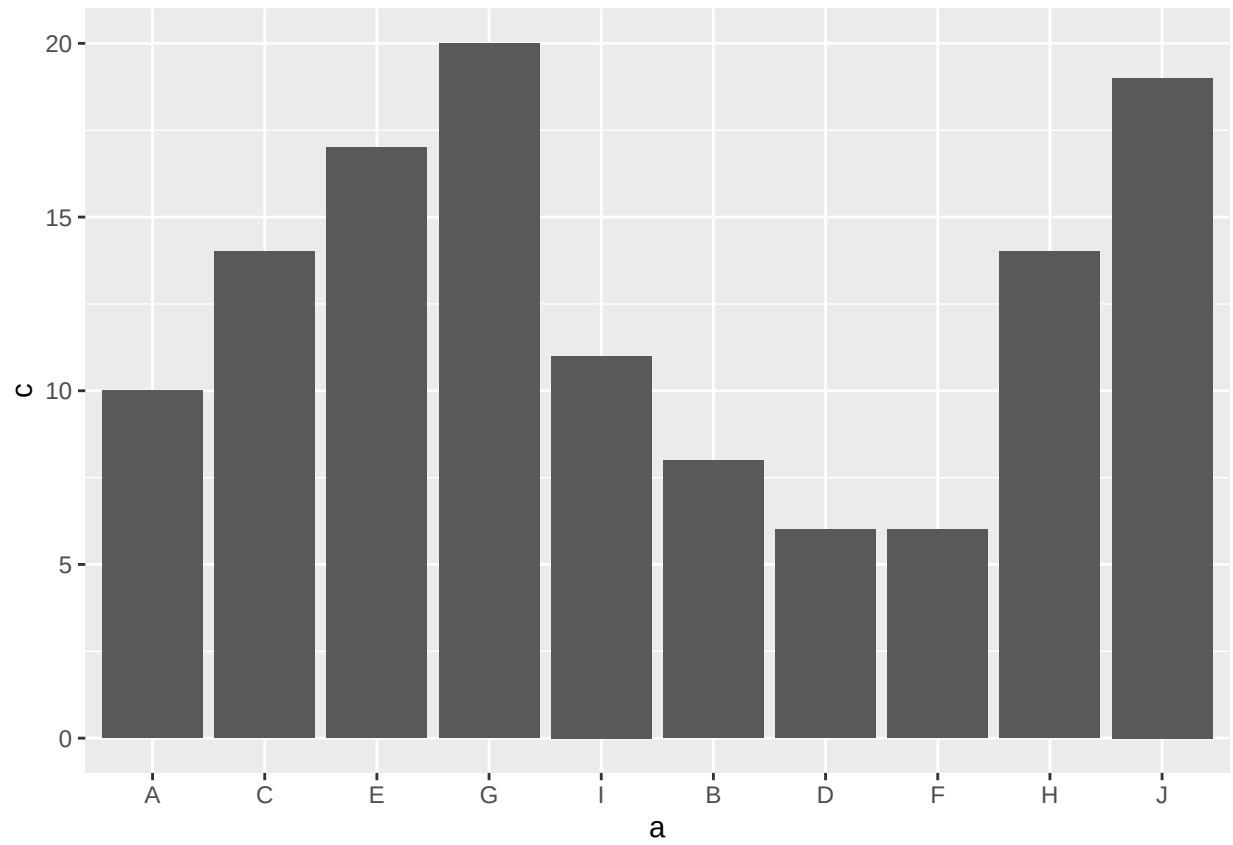


Order Bars

```
a <- c("A", "B", "C", "D", "E", "F", "G", "H", "I", "J")
b <- c(11, 11, 11, 12, 13, 11, 13, 12, 13, 13)
c <- c(10, 8, 14, 6, 17, 6, 20, 14, 11, 19)
d <- data.frame(a, b, c)
d
```

```
##   a  b  c
## 1  A 11 10
## 2  B 11  8
## 3  C 11 14
## 4  D 12  6
## 5  E 13 17
## 6  F 11  6
## 7  G 13 20
## 8  H 12 14
## 9  I 13 11
##10  J 13 19
```

```
ggplot(data = d, aes(x = a, y = c)) + geom_bar(stat = "identity") +
  scale_x_discrete(limits = c("A", "C", "E", "G", "I", "B", "D", "F", "H", "J"))
```



```
ggplot(data = d, aes(x = reorder(a, b), y = c)) + geom_bar(stat = "identity")
```

